



AI Research Group

CENTRE FOR ADVANCED STUDIES

Dr. A.P.J. Abdul Kalam Technical University
Lucknow



Research Seminar Series

Seminar -3

Deep Learning in Medical Imaging: Automated Detection of COVID-19 from Chest X-Rays using deep convolutional neural network

DATE & TIME:

**22nd January 2021,
4:00 PM.**

PLATFORM & WEB-LINK

**Google Meet :
<https://meet.google.com/odi-ouba-wgz>**



SPEAKER

Saumya Yadav

**Junior Research Fellow,
Centre for Advanced
Studies (Till Dec, 2020)**



ABSTRACT

The lethal novel coronavirus disease 2019 (COVID-19) pandemic is affecting the health of the global population severely. There is a need for effective and reliable systems that perform automatic detection and mass screening of COVID-19 as a quick alternative diagnostic option to control its spread. Deep-learning-based robust system is proposed to detect COVID-19 using chest X-ray images. Deep learning is a part of machine learning which is utilized to solve complex problems with the state-of-the-art performance on computer vision and image processing. It is widely used for medical imaging giving a high performance in segmentation, classification, and detection tasks. Infected patient's chest X-ray images reveal numerous opacities in comparison to healthy lungs images which is used by a deep learning algorithm to generate a model to facilitate an accurate diagnostics for multi-class classification and binary classification. The proposed model performs real-time disease detection in a system equipped with a GPU device and can reduce the workload of radiologists by classifying thousands of images on a single click to generate a probabilistic report in real-time.